S3.1 The periodic table: Classification of elements

Electrons and the periodic table

Learning outcomes

By the end of this section you should be able to:

- Deduce the electron configuration of an atom up to Z = 36 from the position of the element in the periodic table.
- Deduce the position in the periodic table of an atom of its electron configuration.
- Identify the alkali metals, halogens, transition elements and noble gases in the periodic table.

In the previous section, S3.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) we looked at blocks in the periodic table which provided information about the sublevel that valence electrons occupy for an element. Throughout topic S2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) we used the periodic table to make predictions about elements and the valence electrons involved during bonding between elements. Since the periodic table provides useful information about electrons for elements, how might we use it to make predictions about the specific electron configuration of each element?

Let's consider the condensed electron configuration for group 1 metals in **Table 1**. Recall from section S3.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) that group 1 elements are in the s-block, meaning their outermost valence electrons are occupying an s-sublevel. Notice how all of the elements in group 1 have one electron occupying an s-sublevel.

Table 1. Condensed electron configuration for group 1 metals.

Element	Period number	Condensed electron configuration
Lithium	2	[He] $2s^1$
Sodium	3	[Ne] $3s^1$
Potassium	4	[Ar] $4s^1$
Rubidium	5	[Kr] $5s^1$
Caesium	6	???

What would you predict the condensed electron configuration for caesium would be? The noble gas that would represent the core electrons would be Xe and it is in period 6. Therefore, the condensed electron configuration would be [Xe] $6s^1$.

Now, let's consider the condensed electron configuration for group 17 non-metals in **Table 2.** Group 17 elements are in the p-block, meaning their outermost valence electrons are occupying a p-sublevel. Notice how all of the elements in group 17 have seven valence electrons with five electrons occupying a p-sublevel.

Table 2. Condensed electron configuration for group 17 non-metals.

Element	Period number	Condensed electron configuration
Fluorine	2	[He] $2s^2 2p^5$
Chlorine	3	[Ne] $3s^2 3p^5$
Bromine	4	[Ar] $4s^2 3d^{10} 4p^5$
Iodine	5	???

What would you predict the condensed electron configuration for iodine to be?
The noble gas that would represent the core electrons would be Kr and it is in period 5. Therefore, the condensed electron configuration would be [Kr] $5s^2 4d^{10} 5p^5$.

The electron configurations of elements are easily predicted using the periodic table because all elements in a group have a common number of valence electrons and the period number shows the outer energy level that is occupied by electrons.

Using the block method to determine electron configuration

We can use the position of elements in the periodic table to deduce their electron configuration; this is known as the block method. It is an alternative method to the order of filling method you looked at in section S1.3.3–4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/). The block method can be used by following these guidelines:

- Identify the element of interest in the periodic table.
- Identify the period number to deduce the outermost energy level that will be occupied by electrons.
- Starting with hydrogen, move along each row, left to right, top to bottom writing out the electron configuration for each block in a period until you reach your element.

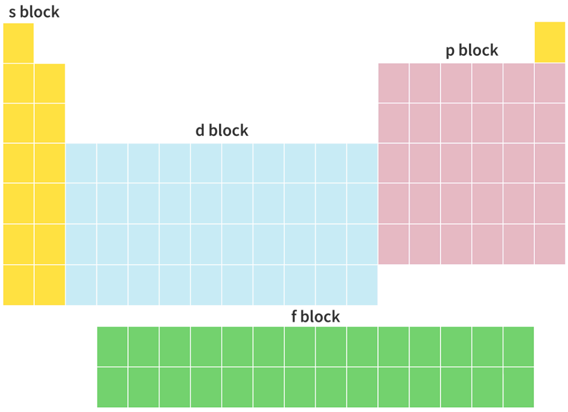
Implementation of these rules are demonstrated in **Interactive 1** for the element arsenic.

Interactive 1. Using the Block Method to Deduce the Electron Configuration for Arsenic.

More information for interactive 1

The slideshow with 10 slides displays the Block Method to deduce the Electron Configuration for Arsenic. The slide and the accompanying text read as follows.
Slide 1. The modern periodic table with a highlighted circle on the element “As”. The text on the left reads, “Identify the element of interest”.
Slide 2. The modern periodic table with a highlighted circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”.
Slide 3. The modern periodic table with the first period highlighted and circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”. The text below the periodic table is $1s^2$.
Slide 4. The modern periodic table with the second period highlighted and circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”. The text below the periodic table is $1s^22s^2$”.
Slide 5. The modern periodic table with the elements from groups 13 to 18 of the second period highlighted and circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”. The text below the periodic table is $1s^22s^22p^6$.
Slide 6. The modern periodic table with the first and second elements of the third period highlighted and a circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”. The text below the periodic table is $1s^22s^22p^63s^2$.
Slide 7. The modern periodic table with the elements of groups 13 to 18 of period 3 highlighted and a circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”. The text below the periodic table is $1s^22s^22p^63s^23p^6$
Slide 8. The modern periodic table with the elements of groups 1 and 2 of period 4 highlighted and a circle on the element “As”. The text on the left reads, Move along each row and write out the electron configuration for each block using the period numbers to guide you. The text below the periodic table is $1s^22s^22p^63s^23p^64s^2$
Slide 9. The modern periodic table with the elements of groups 3 to 12 of period 4 highlighted and a circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you”. The text below the periodic table is $1s^22s^22p^63s^23p^64s^23d^{10}$
Slide 10. The modern periodic table with the elements of groups 13 to 15 of period 4 highlighted and a circle on the element “As”. The text on the left reads, “Move along each row and write out the electron configuration for each block using the period numbers to guide you.” The text below the periodic table is
$1s^22s^22p^63s^23p^64s^23d^{10}4p^3$

Test your understanding of the block method by dragging the electron configurations into their proper locations in the periodic table.



1 $1s^22s^22p^63s^23p^64s^23d^{10}4p^5$

2 $1s^22s^22p^4$

3 $1s^2$

4 $[Xe]6s^24f^4$

5 $1s^22s^22p^63s^23p^64s^23d^7$

6 $[Ne]3s^2$

Check

Interactive 1. Drag and Drop the Electron Configurations Into Their Proper Locations in the Periodic Table.

More information for interactive 1

An interactive periodic table is divided into s-block on the left, p-block on the right, d-block between s- and p-blocks, and f-block positioned at the bottom. The table is organised into rows as periods, from 1 to 7, and columns as groups, from 1 to 18.
Users need to drag and drop electron configurations into their respective locations in the periodic table.
The drop zones are provided in the following positions:
s-block: Group 2, Period 3
d-block: Group 9, Period 4
p-block: Group 16, Period 2
p-block: Group 17, Period 4
s-block: Group 18, Period 1
f-block: Group 3, Period 6
Provided electron configuration options:

1. $1s^22s^22p^63s^23p^64s^23d^{10}4p^5$
2. $1s^22s^22p^4$
3. $1s^2$
4. $[Xe]6s^24f^4$
5. $1s^22s^22p^63s^23p^64s^23d^7$
6. $[Ne]3s^2$

This interactivity helps users understand the relationship between the electronic configuration and the position of elements within different blocks of the periodic table.
Read below for the solution:

s-block: Group 2, Period 3 - $[Ne]3s^2$

d-block: Group 9, Period 4 - $1s^22s^22p^63s^23p^64s^23d^7$

p-block: Group 16, Period 2 - $1s^22s^22p^4$

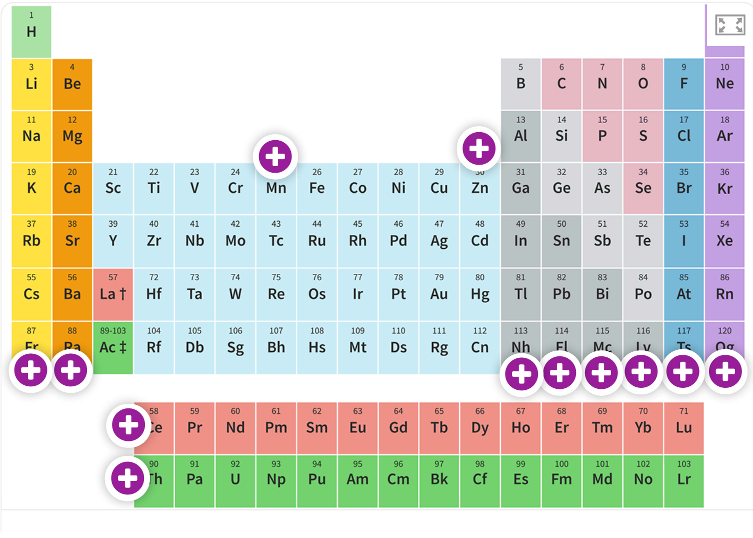
p-block: Group 17, Period 4 - $1s^22s^22p^63s^23p^64s^23d^{10}4p^5$

s-block: Group 18, Period 1 - $1s^2$

f-block: Group 3, Period 6 - $[Xe]6s^24f^4$

Further classification in the periodic table

While groups of elements with similar properties are classified according to group number, there are informal names often used to refer to a group or groups of elements. These informal names often come from historical usage. **Interactive 3** shows a periodic table. Click on each group to see the informal name assigned to that group of elements.



Interactive 3. Informal names for groups in the periodic table.

> **Study skills**
>
> While these informal names are often useful names to reference specific groups in the periodic table, they are not internationally recognised by IUPAC as names for groups of elements. You must be familiar with the classifications of alkali metals (group 1), transition elements (groups 3—12), halogens (group 17) and noble gases (group 18) but are not required to know the others.

Table 3 provides some additional detail about the alkali metals, transition elements, halogens and the noble gases.

Table 3. Additional detail about the alkali metals, transition elements, halogens and the noble gases.

Element Group	Category	Details
Alkali metals	Origins of name	Comes from the tendency of these metals to form alkaline solutions when they react with water.
	Properties of the elements	Soft metals that are very reactive with water and many non-metals.
	Links to further sections	The reactions of the alkali metals with water will be looked at more closely in section S3.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/).

Element Group	Category	Details
Transition elements	Origins of name	These elements show transitionary behaviour intermediate between elements in the s-block and those in the p-block.
	Properties of the elements	Can form ions with different ionic charges. Metals have high melting points and high conductivities.
	Links to further sections	HL will look more closely at formally defining transition elements in section S3.1.8—9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/).
Halogens	Origins of name	Name is Greek and means “salt producing”.
	Properties of the elements	Highly reactive and form ionic salts when they react with metals.
	Links to further sections	The reactions of halogens with halide ions will be looked at more closely in section S3.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/).
Noble gases	Origins of name	Name is German and means “immaculate”.
	Properties of the elements	Colourless monoatomic gases that have little to no reactivity with other elements.
	Links to further sections	Noble gases are often used to simplify electron configurations of other elements since they have an outer energy level that is completely filled. You learned about this condensed electron configuration in section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/).

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate how to plan and initiate a CAS experience
- Demonstrate engagement with issues of global significance

“If you can’t grow it, you have to mine it”. What do you think this expression means? It’s another way of saying that if you cannot grow a material, such as plants used for food or clothing, then you have to mine it (extract it from the ground). According to the USGS (https://elements.visualcapitalist.com/all-the-metals-mined-in-2021/), 2.8 billion tonnes (2.80×10^{12} kg) of metals were mined in 2021 alone. The process of extracting these metals from their ores can have significant negative effects

(https://www.americangeosciences.org/critical-issues/faq/how-can-metal-mining-impact-environment#:~:text=If%20not%20prevented%20or%20controlled,impact%20stream%20habitats%20and%20groundv
the environment.

As an IB student, think about what you can do to help offset the environmental effects of mining. Perhaps you could start by investigating if any recycling programmes such as those used to recycle electrical waste (known as E-waste) or metals exist in your school or local community. Alternatively, you could raise awareness of the environmental damage caused by the mining of metals and other such materials.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication skills — Using terminology consistently and correctly
- **Time required to complete activity:** Approximately 30 minutes
- **Activity type:** Pair activity

Electron configuration battleship game

Materials (per student):

- Two laminated periodic tables. If lamination is not available paper copies will work too.
- One folder.
- One whiteboard marker. If using paper copies of the periodic table a pen or pencil would work.

Periodic table
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

Individual instructions:

- Glue or staple your two periodic tables to the inside of a folder as shown.
- One periodic table will be your gameboard, the other will be your opponents. You will need to keep track of both during this game.
- On your gameboard you will add five boats by completely colouring in the box of the elements that make up the boat. Boats must be made up of adjacent elements (vertical or horizontal). You should have the following boats:
  - One five-element boat
  - Two four-element boats
  - Two three-element boats

Pair instructions:

- Sit opposite your partner on the other side of a table.
- With your partner, connect your folders “laptop style” using a paperclip to join the top. Ensure that you are not able to view the contents of your partner’s folder.
- You will take turns trying to guess the location of your partner’s boats, you are only permitted to use electron configurations to state the position of the element. Be patient with one another, this is for practice.
- Keep track of your guesses for your opponent on the bottom periodic table, and their guesses for your boats on the top periodic table.
- Use an X for misses and colour the square in completely for a hit.
- If needed, use a periodic table such as the one shown to help guide you.

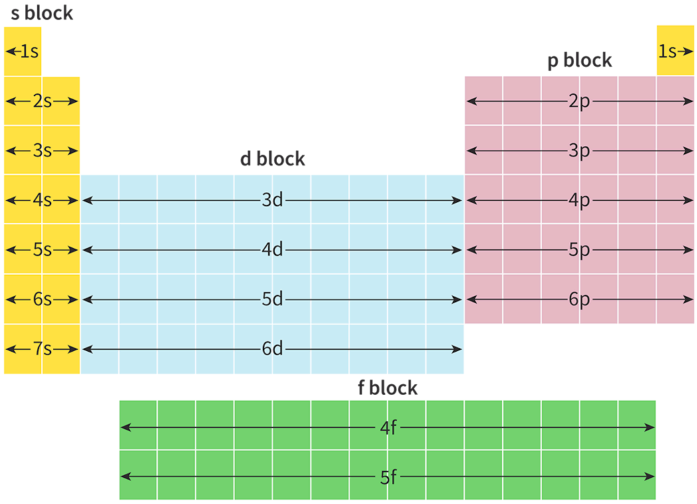


Figure 6. Periodic table blocks.

More information for figure 6

This image shows the periodic table divided into blocks that represent different electron configurations. The left section is highlighted in yellow and labeled as the 's' block, containing arrows pointing horizontally from 1s to 7s. To the right, the blue section indicates the 'd' block, with arrow lines marked from 3d to 6d, running horizontally. On the far right, akin to the 'd' block, is the pink-colored 'p' block, with arrows for 2p to 6p extending horizontally. Below the main table, there is a green section labeled as the 'f' block, showing arrows moving horizontally from 4f to 5f. This layout illustrates the organization of elements based on their electron configurations in atomic orbitals.

[Generated by AI]